

SHILATE: SOFTWARE-DEFINED VEHICLE REINFORCEMENT LEARNING — FROM SIMULATION TO SDV DEPLOYMENT

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ABSTRACT

The proliferation of software-defined vehicle (SDV) platforms has opened an opportunity to combine modern deep reinforcement learning (RL) with real automotive runtime stacks. This thesis presents **SHILATE** — a complete, open-source pipeline that trains a Proximal Policy Optimization (PPO) agent inside a Unity physics simulation and deploys the resulting policy onto an Eclipse Leda SDV stack, bridging the gap between laboratory RL research and realistic vehicle software deployment.

The simulation component is built in Unity 2022 LTS. A single rear-wheel-drive vehicle navigates a circular obstacle course instrumented with a 21-ray LiDAR-style sensor. The training bridge computes a shaped reward signal from progress, collision events, and episode milestones, and communicates with the Python training process exclusively over MQTT, keeping the two subsystems fully decoupled. A dedicated Editor window (*SHILATE Training Controller*) provides real-time health monitoring, metric graphs, and one-click training control without leaving the Unity Editor.

The Python training stack wraps the simulation in a Gymnasium-compliant environment, configures Stable-Baselines3 PPO with a two-layer actor-critic network, and streams structured telemetry back to the Editor. Trained checkpoints are subsequently deployed to the Eclipse Leda SDV image via a containerised MQTT-to-Kuksa bridge and a Velocitas vehicle application that enforces operational limits (overspeed, RPM redline) on the running agent.

Experimental results demonstrate that the PPO agent converges reliably on the obstacle course within 100 000 interaction steps and that the end-to-end deployment chain correctly propagates vehicle telemetry from the simulation through the VSS data layer to the Velocitas safety monitor. The project shows that an integrated sim-to-SDV pipeline is achievable with commodity open-source tooling, and identifies curriculum learning and multi-environment parallelisation as natural extensions for future work.

Keywords: reinforcement learning, software-defined vehicle, Unity simulation, PPO, MQTT, Eclipse Leda, Kuksa, Stable-Baselines3, Gymnasium.

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1. INTRODUCTION

1.1 L^AT_EX INSTRUCTIONS

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2. BIBLIOGRAPHY

- [1] A. Einstein, “Zur Elektrodynamik bewegter Körper. (German) [On the electrodynamics of moving bodies],” *Annalen der Physik*, vol. 322, no. 10, pp. 891–921, 1905. DOI: <http://dx.doi.org/10.1002/andp.19053221004>.

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